

Science

Nature Notebook
Nature Walks & Scouting
Botany Lessons
Botany Labs





About the Course

This course includes the following topic(s): Botany Lessons, Botany Labs, Nature Notebook: Grades 9-12, Nature Walks & Scouting: Grades 9-12

About Nature Notebook: Grades 9-12

Outdoor work is established or continued as a lifelong habit. Optional resources are provided in science lessons and on the Alveary bookshelf.

About Nature Walks & Scouting: Grades 9-12

Outdoor work is established or continued as a lifelong habit. Optional resources are provided in science lessons and on the Alveary bookshelf.

About Botany Lessons

An elective in Life Science, Botany guides learners through the basic foundations of the discipline, while incorporating historical context, modern developments, current events, and citizenship. This course provides guidance and flexibility for teachers and learners to adjust the course for personal needs and preferences.

About Botany Labs

Note that labs are an essential part of science in which students engage with the Things they are reading about and practice the scientific method.



Placement & Combining Tips

Nature Notebook: Grades 9-12

Learners may be combined and follow their own interests.

Nature Walks & Scouting: Grades 9-12

Learners may follow their own interests or follow the plan of their local scouting troop or natural history club.

Botany Lessons

There are no specific prerequisites or concurrents for this course.



Scheduling

GRADE	SCHEDULE INFO.	BOOKS
9-12	Nature Notebook: Grades 9-12 1+ time/week 20 min+	
9-12	Nature Walks & Scouting: Grades 9-12 1 time/week 30 min+	
9-12	Botany Lessons 5 times/week 45 min	A Peterson Field Guide To Western Trees Botany in a Day Growing a Revolution: Bringing Our Soil Back to Life Pacific Coast Tree Finder Rocky Mountain Tree Finder Shanley's Quest Card Game The Milkweed Lands: An Epic Story of One Plant: Its Nature and Ecology Oaxaca Journal Tree Finder - Eastern US What a Plant Knows

GRADE	SCHEDULE INFO.	BOOKS
		Winter Tree Finder (Eastern US) The Hidden Life of Trees: The Illustrated Edition The Sibley Guide to Trees Elementary Studies in Plant Life
9-12	Botany Labs 1 time/week 60 min	

Sample Weekly View

Day 1	Day 2	Day 3	Day 4	Day 5
Science: Botany				
Botany Lessons Nature Walks & Scouting: Grades 9-12	Botany Lessons	Botany Lessons	Botany Lessons Nature Notebook: Grades 9-12	Botany Lessons Botany Labs



Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

LINKS: Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

Botany Lessons

- ☐ Select a science book from the Alveary bookshelf for personal reading time, as appropriate.
- ☐ Bookmark or print Quick Links, as needed.

Term Prep & Teacher Tips

Botany Lessons

- ☐ Gather household items, typically easy for students to scavenge or teachers to obtain locally:
 - various plants specimens, found or obtained as detailed in Elementary Plant Studies
 - lily flower (Term 1)
 - found twigs and fruits/seeds
 - needle and/or blade for dissecting
 - forceps/tweezers for dissecting
 - large, transparent cup/glass
 - masking tape
 - aluminum foil
 - 2 glass jars
 - salt
 - oil
 - microscope coverslips and slides
 - paper towels
 - saucer
 - recycled box
 - pins
 - cotton ball
 - various corks
 - recycled measuring spoon/cup marked in tsp/Tbsp or mL
 - recycled apple corer or trowel
 - place to dig soil
 - distilled water
 - zip top sandwich bags for soil samples
 - small jars with secure lid for each soil sample
 - dropper
 - pea/bean seeds
 - potting materials

- denatured alcohol
- iodine
- potato
- matches
- candle
- red food coloring/dye
- straw
- materials by student choice
- optional: tulip bulb

Reminders

Botany Lessons

☐ Note that Lab in week 7 requires a lily flower. Make a calendar note to obtain this closer to the appropriate time.



Books & Resources

For book rationales and purchase options, click the Book List link or scan the QR code below.

∞ [View Book List Details](#)

Science: Botany

Botany Lessons



A Peterson Field Guide To Western Trees



Botany in a Day



Growing a Revolution: Bringing Our Soil Back to Life



Pacific Coast Tree Finder



Rocky Mountain Tree Finder



Shanley's Quest Card Game



The Milkweed Lands: An Epic Story of One Plant: Its Nature and Ecology



Oaxaca Journal



Tree Finder - Eastern US



What a Plant Knows



Winter Tree Finder (Eastern US)



The Hidden Life of Trees: The Illustrated Edition



The Sibley Guide to Trees



Elementary Studies in Plant Life



Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

∞ [View Basic Supplies](#)

∞ [View Supply List Details](#)

Science: Botany



Calcium Hydroxide



Cobalt Chloride



Glass Rod



Household Items - Science: Botany



Calcium Carbonate



Prepared Microscope Slides



Student Microscope

Botany Labs



Slides and Coverslips



Quick Links

Science: Botany

Botany Lessons

∞ [Seek app from iNaturalist](#)

∞ [Nature Journal Connection](#)

∞ [Alveary Bookshelf](#)

∞ [Elementary Studies in Plant Life](#)

∞ [Elementary Studies in Plant Life Annotations](#)

∞ [AmScope key for slide IDs](#)

[Click THIS text or scan the QR code for links.](#)



Science: Botany

How To Approach



Introduce

- If starting a new book or a new topic in the book, then look at the title or a picture and take a moment to consider previous ideas and experiences.
- If continuing a previous reading, recap what was read previously. Often, the title of the book's section can help to draw out the main idea.



Read

- Read or do, as instructed in the lessons, making note of or flagging unfamiliar terms, interesting ideas, important dates, inspiring quotes, etc.
- Use supportive strategies and educational tools to reduce frustration and better engage the mind, as appropriate. These could include, but are not limited to, the use of eBooks, pictures, audio, read-aloud, buddy reading, colored reading strips, etc.
- As they read, learners record ideas in a notebook or binder using outlines, diagrams, graphic organizers, or other methods (or a combination of methods) that suit them. These recordings can be a helpful mechanism for remembering or a mini-narration to support understanding.
- If learners do not understand a word or concept, do not worry. Try reading over the passage again, studying a picture or diagram, connecting the idea to something from real life, or practicing chapter exercises. The lab/field work further supports major concepts from the text.



Narrate

- Process the ideas of the lesson by retelling, defining a concept, explaining the links in a chain of thought, etc. Do this orally or silently to yourself.
- Use words, pictures, outlines, etc.
- If a particular idea cannot be narrated, then read or examine the text again.



Discuss

- Consider with the teacher any thoughts, confusion, or concerns about the passage.
- If understanding is still uncertain, try rereading the passage or do some personal research on the topic.



Connect

- Follow any extra links, examine any sidebars in the text, or pursue additional reading, depending on student interest.

Science: Botany

How To Approach



Introduce

- Regardless of how many days are required to complete a particular activity, every Science Lab has the same flow, which follows the scientific method.
- Relevant concept(s) are introduced in the text.
- Your notebook entry begins with the introductory/prelab narration, including relevant information that you have read or previously experienced, what you plan to do in the lab, and any hypothesis or anticipated result.



Lab Procedure

- Perform the procedure according to the instruction, recording in your notebook what you do as it happens. This can be a challenge, but is an extremely important skill.
- Record all data and observations in the lab notebook.



Analysis & Conclusions

- After all data is collected, analyze the results by considering how the data reflects the introduced concepts and whether the hypothesis is supported by the data.
- If the data and observations do not support the hypothesis, reflect on why and what further testing would be interesting or helpful.



Term 1

WEEK 1 45m Botany Lessons - Lesson 1

Botany Literature

Materials: What a Plant Knows

→ INTRO

At least one day each week will be spent reading important works of botanical literature. The lesson plans give you an idea of how many pages to read each week to finish the book by the end of the term. If one day is not enough, you might read about half on one day and half on the flex day. If one day is enough, then decide how you would like to use your flex day. Consider the ideas provided on Day 4 of the week.

→ READ, NARRATE, & DISCUSS

Read about 14 pages per week; p.3-17.

→ WRITE

Write a narration in your Botany Notebook.

• PLAN WEEKLY

- ☐ Nature walk - Look for the plants and/or features mentioned in your books. Record observations.
- ☐ science free read

WEEK 1 45m Botany Lessons - Lesson 2

How Plants Work, Life Cycle of Flowering Plants

Materials: Elementary Studies in Plant Life

→ READ, NARRATE, & DISCUSS

Ch.1. Work through the material, adding drawings to your notebook. You may need to visit local nurseries, botanical gardens, neighbors' yards, and gardens to find specimens to examine. Pick a question at the end to answer fully in your notebook for your narration.

WEEK 1 45m Botany Lessons - Lesson 3

Identification

Materials: Botany in a Day

→ READ, NARRATE, & DISCUSS

Forward, Region Covered, and p.1-9

→ NOTE

You may or may not agree with the author's presentation of the evolution of plants. Still, it is important to know current botanical thinking, especially as we seek to understand how plants have been organized. This will help us as we begin to learn plant families.

WEEK 1 45m Botany Lessons - Lesson 4

Flex Day

Materials: Selected botany literature OR lab/field supplies

→ INTRO

Use this flex day to finish the week's reading from What a Plant Knows, or continue any lab/field activities. Suppose you have completed the week's activities. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as Growing a Revolution, or more active by pursuing a special study of interest.



Term 1

WEEK 1 45m Botany Lessons - Lesson 5

Special Study

Materials: The Milkweed Lands

→ INTRO

Notice how the author and illustrator make a study of and learn so much about this one plant. Whether you have milkweed near you and can notice their observations, or you make a similar study of a particular plant in your locale, enjoy their method as you develop your own.

→ READ, NARRATE, & DISCUSS

Half of Ch.1 (to the end of "Dangerous Colors" p.8)

WEEK 1 60m Botany Labs - Lesson 1

Materials: Elementary Studies in Plant Life

Prep: Read Student Tip.

→ OBSERVE

Notice a variety of plants this week, doing the practical work from Ch.1. Notice root and shoot, and how the leaves and branches are attached in the different cases.

Notice plants in which flowers are fading, and fruit is ripening in their place. Some activities may be delayed depending on the time of year and your geographic location. Do what you can to gain the most from the chapter. If the plants or trees mentioned are unavailable locally, look up the item online. Also, you may need to visit local nurseries, botanical gardens, and neighbors' yards and gardens to find specimens to examine.

★ STUDENT TIP

Labs and prompts for your field work are included in your lesson plans, but it is important to establish a personal habit of outdoor walks as a part of life!

WEEK 2 45m Botany Lessons - Lesson 6

Botany Literature

Materials: What a Plant Knows

→ READ, NARRATE, & DISCUSS

Read about 14 pages per week; p.17-31.

→ WRITE

Write a narration in your Botany Notebook.

• PLAN WEEKLY

Nature walk - Look for the plants and/or features mentioned in your books. Record observations.
 science free read

WEEK 2 45m Botany Lessons - Lesson 7

How Plants Work, Use of Water

Materials: Elementary Studies in Plant Life

→ READ, NARRATE, & DISCUSS

Ch.2

WEEK 2 45m Botany Lessons - Lesson 8

Identification

Materials: Botany in a Day

→ READ, NARRATE, & DISCUSS

p.10-18



Term 1

WEEK 2 45m Botany Lessons - Lesson 9

Flex Day

Materials: Selected botany literature OR lab/field supplies

→ NOTE

Use this flex day to finish the week's reading from What a Plant Knows, or continue any lab/field activities. Suppose you have completed these activities for the week. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as Growing a Revolution, or more active by pursuing a special study of interest.

WEEK 2 45m Botany Lessons - Lesson 10

Special Study

Materials: The Milkweed Lands

→ READ, NARRATE, & DISCUSS
Finish Ch.1 from p.8.

WEEK 2 60m Botany Labs - Lesson 2

Materials: Elementary Studies in Plant Life

→ LAB DAY
Perform the practical work from Ch.2. Record your work in your notebook.

WEEK 3 45m Botany Lessons - Lesson 11

Botany Literature

Materials: What a Plant Knows

→ READ, NARRATE, & DISCUSS
Read about 14 pages per week; p.31-45.

→ WRITE
Write a narration in your Botany Notebook.

• PLAN WEEKLY

☐ Nature walk - Look for the plants and/or features mentioned in your books. Record observations.
☐ science free read

WEEK 3 45m Botany Lessons - Lesson 12

How Plants Work, Soil Science

Materials: Links

→ INTRO

Scientists now know that soil science is fundamentally important to plant growth and even human health. This is because of the vast number of organisms that live in the soil food web. Look at a basic diagram of this web at the link and copy it into your Botany Notebook. Then, watch the videos to consider how it works and why it's important.

→ VIEW

∞ Image Link: Soil Food Web

→ VIEW, NARRATE, & DISCUSS

∞ Video Link: The Living Soil Beneath Our Feet (2:49)

∞ Video Link: How Regenerative Agriculture Brings Life Back (15:22)



Term 1

WEEK 3 45m Botany Lessons - Lesson 13

Identification

Materials: Botany in a Day, Shanleya's Quest card game

→ READ, NARRATE, & DISCUSS
p.19-21

→ VIEW

Watch the author explain further.

∞ Link: Tutorial

∞ Link: Written Explanation in larger print from the website

→ PLAY

Find a friend and begin playing the card games to familiarize yourself with the families and recognize the patterns.

WEEK 3 45m Botany Lessons - Lesson 14

Flex Day

Materials: Selected botany literature OR lab/field supplies

→ NOTE

Use this flex day to finish the week's reading from What a Plant Knows or continue any lab/field activities. Suppose you have completed these activities for the week. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as Growing a Revolution, or more active by pursuing a special study of interest.

WEEK 3 45m Botany Lessons - Lesson 15

Special Study

Materials: The Milkweed Lands

→ READ, NARRATE, & DISCUSS
Half of Ch.2 (to p.27)

WEEK 3 60m Botany Labs - Lesson 3

Materials: Link, household items indicated, microscope, and glass slides

Prep: Read Student Tip.

→ LAB DAY

Use the procedure at the link to evaluate microbial life in soil from at least 3 locations. The locations may be at the same site, but they should be some distance apart. Soil that is even 10-20 yards apart may be quite different. It is okay to be unsure of what you are viewing under the microscope.

→ VIEW

∞ Link: Soil Sampling Procedure

★ STUDENT TIP

Plan to go to another type of habitat next week. If you collected soil from a neighborhood this week, perhaps go to a forest or a pond next week.



Term 1

WEEK 4 45m Botany Lessons - Lesson 16

Botany Literature

Materials: What a Plant Knows

→ READ, NARRATE, & DISCUSS

Read about 14 pages per week; p.46-59.

→ WRITE

Write a narration in your Botany Notebook.

• PLAN WEEKLY

Nature walk - Look for the plants and/or features mentioned in your books. Record observations.

science free read

WEEK 4 45m Botany Lessons - Lesson 17

How Plants Work, More Soil Science

Materials: Links

→ READ, NARRATE, & DISCUSS

∞ Article Link: Soil Health

→ VIEW

∞ Video Link: Hopi Agriculture (4:12)

∞ Video Link: Indigenous Farmer Solving Food Insecurity (16:11)

∞ Video Link: Tiny Hooves Rewilding the Great Plains (13:25)

WEEK 4 45m Botany Lessons - Lesson 18

Identification

Materials: Botany in a Day, Shanleya's Quest card game

→ READ, NARRATE, & DISCUSS

p.79, 91, 109, 157, 163, 179, 192, and 207

→ PLAY

Continue playing the card games until you are confident in your ability to recognize the 8 families.

WEEK 4 45m Botany Lessons - Lesson 19

Flex Day

Materials: Selected botany literature OR lab/field supplies

→ NOTE

Use this flex day to finish the week's reading from What a Plant Knows or continue any lab/field activities. Suppose these activities are finished for the week. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as Growing a Revolution, or more active by pursuing a special study of interest.

WEEK 4 45m Botany Lessons - Lesson 20

Special Study

Materials: The Milkweed Lands

→ READ, NARRATE, & DISCUSS

Finish Ch.2 from p.28.



Term 1

WEEK 4 60m Botany Labs - Lesson 4

Materials: Link, household items indicated, microscope, and glass slides

→ LAB DAY

This week, go to a different habitat. Use the procedure at the link to evaluate microbial life in soil from at least 3 locations.

→ VIEW

∞ Link: Soil Sampling Procedure

WEEK 5 45m Botany Lessons - Lesson 21

Botany Literature

Materials: What a Plant Knows

→ READ, NARRATE, & DISCUSS

Read about 14 pages per week; p.60-73.

→ WRITE

Write a narration in your Botany Notebook.

• PLAN WEEKLY

Nature walk - Look for the plants and/or features mentioned in your books. Record observations.

science free read

WEEK 5 45m Botany Lessons - Lesson 22

How Plants Work, Photosynthesis Basics

Materials: Elementary Studies in Plant Life

→ READ, NARRATE, & DISCUSS

Ch.3

WEEK 5 45m Botany Lessons - Lesson 23

Identification

Materials: Botany in a Day, Shanley's Quest card game

→ PRACTICE

Practice recognizing the 8 main families by playing the card games and observing plants. You will be introduced to a couple of new families each week. It might be possible to find all of them in the fall in your area. Ask local botanists at gardens, nature museums, or nature trails to assist you in your search. Read about and search for members of the Nymphaeaceae (p.50) and Lauraceae (p.53) families.

WEEK 5 45m Botany Lessons - Lesson 24

Flex Day

Materials: Selected botany literature OR lab/field supplies

→ NOTE

Use this flex day to finish the week's reading from What a Plant Knows or continue any lab/field activities. Suppose these activities are finished for the week. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as Growing a Revolution, or more active by pursuing a special study of interest.



Term 1

WEEK 5 ☐ 45m Botany Lessons - Lesson 25

Special Study

☐ Materials: The Milkweed Lands

→ READ, NARRATE, & DISCUSS
Half of Ch.3 (to the end of p.47)

WEEK 5 ☐ 60m Botany Labs - Lesson 5

☐ Materials: Elementary Studies in Plant Life, pea/bean seeds, and potting materials

→ LAB DAY
Perform the practical work from Ch.3. Record your work in your notebook. Also, take a few minutes to start some pea or bean seeds for Week 10. Be sure to keep them watered!

WEEK 6 ☐ 45m Botany Lessons - Lesson 26

Botany Literature

☐ Materials: What a Plant Knows

→ READ, NARRATE, & DISCUSS
Read about 14 pages per week; p.73-86.

→ WRITE
Write a narration in your Botany Notebook.

• PLAN WEEKLY

☐ Nature walk - Look for the plants and/or features mentioned in your books. Record observations. Begin collecting twigs and seeds for use in Term 2.
☐ science free read

WEEK 6 ☐ 45m Botany Lessons - Lesson 27

How Plants Work, More on Photosynthesis

☐ Materials: Elementary Studies in Plant Life

→ READ, NARRATE, & DISCUSS
Ch.17

→ VIEW
∞ Video Link: Photosynthesis Video (13:37)

WEEK 6 ☐ 45m Botany Lessons - Lesson 28

Identification

☐ Materials: Botany in a Day

→ READ
Read about and search for members of the Stonecrop (p.64) and Vitaceae (p.67) families.

WEEK 6 ☐ 45m Botany Lessons - Lesson 29

Flex Day

☐ Materials: Selected botany literature OR lab/field supplies

→ NOTE
Use this flex day to finish the week's reading from What a Plant Knows or continue any lab/field activities. Suppose these activities are finished for the week. In that



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case, you may use this day to tailor the course to be more literary by reading an additional text, such as *Growing a Revolution*, or more active by pursuing a special study of interest.

WEEK 6 45m Botany Lessons - Lesson 30

Special Study

Materials: The Milkweed Lands

→ READ, NARRATE, & DISCUSS
Finish Ch.3 (from p.48).

WEEK 6 60m Botany Labs - Lesson 6

Materials: Elementary Studies in Plant Life

→ LAB DAY
Perform a selection of the practical work from Ch.17. Record your work in your notebook.

WEEK 7 45m Botany Lessons - Lesson 31

Botany Literature

Materials: What a Plant Knows

→ READ, NARRATE, & DISCUSS
Read about 14 pages per week; p.86-102.

→ WRITE
Write a narration in your Botany Notebook.

• PLAN WEEKLY

☐ Nature walk - Look for the plants and/or features mentioned in your books. Record observations. Collect twigs and seeds for use in Term 2.
☐ science free read

WEEK 7 45m Botany Lessons - Lesson 32

How Plants Work, Flower Anatomy

Materials: Elementary Studies in Plant Life

→ READ, NARRATE, & DISCUSS
Ch.4

WEEK 7 45m Botany Lessons - Lesson 33

Identification

Materials: Botany in a Day

→ READ
Read about and search for members of the Lythraceae (p.69) and the Onagraceae (p.70) families.

WEEK 7 45m Botany Lessons - Lesson 34

Flex Day

Materials: Selected botany literature OR lab/field supplies

→ NOTE
Use this flex day to finish the week's reading from *What a Plant Knows* or continue any lab/field activities. Suppose these activities are finished for the week. In that



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case, you may use this day to tailor the course to be more literary by reading an additional text, such as *Growing a Revolution*, or more active by pursuing a special study of interest.

WEEK 7 45m Botany Lessons - Lesson 35

Special Study

Materials: The Milkweed Lands

→ READ, NARRATE, & DISCUSS
Half of Ch.4 (to the end of p.75)

WEEK 7 60m Botany Labs - Lesson 7

Materials: Link, at least one lily, and a dissecting kit (supplies noted in lab)

→ DISSECTION

Procure a lily flower. You might have to purchase one at a florist or local grocery store. Read the guide linked below, and then dissect your flower. Sketch and label all the parts using the dissection guide. Have a magnifying lens handy to take a close-up look.

→ VIEW

∞ Link: Dissection Guide

WEEK 8 45m Botany Lessons - Lesson 36

Botany Literature

Materials: What a Plant Knows

→ READ, NARRATE, & DISCUSS
Read about 14 pages per week; p.102-116.

→ WRITE

Write a narration in your Botany Notebook.

• PLAN WEEKLY

☐ Nature walk - Look for the plants and/or features mentioned in your books. Record observations. Collect twigs and seeds for use in Term 2.

☐ science free read

WEEK 8 45m Botany Lessons - Lesson 37

How Plants Work, Plant Forms

Materials: Elementary Studies in Plant Life

→ READ, NARRATE, & DISCUSS
Ch.5

WEEK 8 45m Botany Lessons - Lesson 38

Identification

Materials: Botany in a Day

→ READ

Read about and search for members of the Ulmaceae (p.87) and Cucurbitaceae (p.98) families.

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WEEK 8 45m Botany Lessons - Lesson 39

Flex Day

Materials: Selected botany literature OR lab/field supplies

→ NOTE

Use this flex day to finish the week's reading from What a Plant Knows or continue any lab/field activities. Suppose these activities are finished for the week. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as Growing a Revolution, or more active by pursuing a special study of interest.

WEEK 8 45m Botany Lessons - Lesson 40

Special Study

Materials: The Milkweed Lands

→ READ, NARRATE, & DISCUSS
Finish Ch.4 (from p.76).

WEEK 8 60m Botany Labs - Lesson 8

Materials: Elementary Studies in Plant Life, pea/bean seeds, and potting materials

→ LAB DAY

Perform the practical work from Ch.5. Record your work in your notebook. Also, take a few minutes to start some pea or bean seeds for Week 10. Be sure to keep them watered!

WEEK 9 45m Botany Lessons - Lesson 41

Botany Literature

Materials: What a Plant Knows

→ READ, NARRATE, & DISCUSS
Read about 14 pages per week; p.116-129.

→ WRITE
Write a narration in your Botany Notebook.

• PLAN WEEKLY

☐ Nature walk - Look for the plants and/or features mentioned in your books. Record observations. Collect twigs and seeds for use in Term 2.
☐ science free read

WEEK 9 45m Botany Lessons - Lesson 42

How Plants Work, Leaf Anatomy

Materials: Elementary Studies in Plant Life, microscope, and prepared slides

→ READ, NARRATE, & DISCUSS
Ch.6

• Using the prepared slide set, examine Slide 12, the Hydrilla verticillata whole mount. Notice the form of the leaf, the features along the margin, the tiny cells that repeat over and over, and anything else that stands out. You will learn more about cells in General Biology. Also, examine Slide 1, the pine leaf cross-section. A cross-section is a slice perpendicular to the length. Find the outer layer, called the epidermis, and the tiny pores through the epidermis that allow the leaf to breathe, called the stomata. Record what you see in your notebook.



Term 1

WEEK 9 45m Botany Lessons - Lesson 43

Identification

Materials: Botany in a Day

→ READ

Read about and search for members of the Cactaceae (p.123) and Ericaceae (p.132) families.

WEEK 9 45m Botany Lessons - Lesson 44

Flex Day

Materials: Selected botany literature OR lab/field supplies

→ NOTE

Use this flex day to finish the week's reading from What a Plant Knows or continue any lab/field activities. Suppose these activities are finished for the week. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as Growing a Revolution, or more active by pursuing a special study of interest.

WEEK 9 45m Botany Lessons - Lesson 45

Special Study

Materials: The Milkweed Lands

→ READ, NARRATE, & DISCUSS

Half of Ch.5 (to p.99)

WEEK 9 60m Botany Labs - Lesson 9

Materials: Elementary Studies in Plant Life

→ LAB DAY

Perform the practical work from Ch.6. Record your work in your notebook.

WEEK 10 45m Botany Lessons - Lesson 46

Botany Literature

Materials: What a Plant Knows

→ READ, NARRATE, & DISCUSS

Read about 14 pages per week; p.129-143.

→ WRITE

Write a narration in your Botany Notebook.

• PLAN WEEKLY

☐ Nature walk - Look for the plants and/or features mentioned in your books. Record observations. Collect twigs and seeds for use in Term 2.

☐ science free read

WEEK 10 45m Botany Lessons - Lesson 47

How Plants Work, Root Anatomy and Physiology

Materials: Elementary Studies in Plant Life

→ READ, NARRATE, & DISCUSS

Ch.7

Term 1

WEEK 10 ☐ 45m Botany Lessons - Lesson 48

Identification

☐ Materials: Botany in a Day

→ READ

Read about and search for members of the Apocynaceae (p.140) and Convolvulaceae (p.145) families.

WEEK 10 ☐ 45m Botany Lessons - Lesson 49

Flex Day

☐ Materials: Selected botany literature OR lab/field supplies

→ NOTE

Use this flex day to finish the week's reading from What a Plant Knows or continue any lab/field activities. Suppose these activities are finished for the week. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as Growing a Revolution, or more active by pursuing a special study of interest.

WEEK 10 ☐ 45m Botany Lessons - Lesson 50

Special Study

☐ Materials: The Milkweed Lands

→ READ, NARRATE, & DISCUSS

Finish the book (from p.100).

WEEK 10 ☐ 60m Botany Labs - Lesson 10

☐ Materials: Elementary Studies in Plant Life

→ LAB DAY

Perform the practical work from Ch.7. Record your work in your notebook.

WEEK 11 ☐ 45m Botany Lessons - Lesson 51

Botany Literature

☐ Materials: What a Plant Knows

→ READ, NARRATE, & DISCUSS

Read about 14 pages per week; p.144-163.

→ WRITE

Write a narration in your Botany Notebook.

• PLAN WEEKLY

☐ Nature walk - Look for the plants and/or features mentioned in your books. Record observations. Collect twigs and seeds for use in Term 2.

☐ science free read

WEEK 11 ☐ 45m Botany Lessons - Lesson 52

How Plants Work, Trees and Shrubs in Winter

☐ Materials: Elementary Studies in Plant Life

→ READ, NARRATE, & DISCUSS

Ch.8



Term 1

WEEK 11 45m Botany Lessons - Lesson 53

Identification

Materials: Botany in a Day

→ READ

Read about and search for members of the Solanaceae (p.146) and Scrophulariaceae (p.150) families.

★ TEACHER NOTE

You may want to have a few different types of flowers for the first week of Term 3! Pick some up from the florist that week, if needed.

WEEK 11 45m Botany Lessons - Lesson 54

Flex Day

Materials: Selected botany literature OR lab/field supplies

→ NOTE

Use this flex day to finish the week's reading from What a Plant Knows or continue any lab/field activities. Suppose these activities are finished for the week. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as Growing a Revolution, or more active by pursuing a special study of interest.

WEEK 11 45m Botany Lessons - Lesson 55

Special Study

Materials: Article Link

→ READ, NARRATE, & DISCUSS

The naturalist's work is never done! We continually learn how much we don't know about plants, animals, and their relationships. Enjoy the article in the link!

∞ Article Link: South Carolina Swamps Harbor a Startling Secret

WEEK 11 60m Botany Labs - Lesson 11

Materials: Elementary Studies in Plant Life

→ LAB DAY

Perform the practical work from Ch.8. Record your work in your notebook.

WEEK 12 45m Botany Lessons - Lesson 56

Exam Week

→ EXAMS

Answer question(s) related to the course.

WEEK 12 45m Botany Lessons - Lesson 57

Exam Week

→ EXAMS

Answer question(s) related to the course.

WEEK 12 45m Botany Lessons - Lesson 58

Exam Week

→ EXAMS

Answer question(s) related to the course.

★ TEACHER NOTE

You may want to have a few different types of flowers for the first week of Term 3! Pick some up from the florist that week, if needed.



Term 1	
WEEK 12 45m Botany Lessons - Lesson 59 Exam Week → EXAMS Answer question(s) related to the course.	
WEEK 12 45m Botany Lessons - Lesson 60 Exam Week → EXAMS Answer question(s) related to the course.	
WEEK 12 60m Botany Labs - Lesson 12 Exam Week → EXAMS Answer question(s) related to the course.	

Term 2

WEEK 1 45m Botany Lessons - Lesson 61

Botany Literature

 Materials: The Hidden Life of Trees: The Illustrated Edition

→ INTRO

To ensure that the beautiful photography in this book is inspiring rather than distracting, take a few moments to look at the pictures on the pages you will read before beginning to read. The lesson plans give you an idea of how many pages to read each week to finish the book by the end of the term. If one day is not enough, then you might read about half on one day and half on the flex day. If one day is enough, then decide how you would like to use your flex day. Consider the ideas provided on Day 4 of the week.

→ READ, NARRATE, & DISCUSS

Read about 14 pages per week; p.1-12.

→ WRITE

Write a narration in your Botany Notebook.

• PLAN WEEKLY

-  Nature walk - look for the plants and/or features mentioned in your botany books, record observations; collect twigs and seeds
-  science free read

WEEK 1 45m Botany Lessons - Lesson 62

How Plants Work, Herbs in Winter

 Materials: Elementary Studies in Plant Life

→ READ, NARRATE, & DISCUSS

Ch.9

WEEK 1 45m Botany Lessons - Lesson 63

Identification

 Materials: Botany in a Day, a few different types of flowers

→ READ, NARRATE, & DISCUSS

p.22-25. Begin to familiarize yourself with the keys on p.26-35 (you will have another week to do this). You may find it helpful to have a few different types of flowers to look at while examining the keys. The goal is to understand how they work and why we use them, rather than to memorize information.

→ VIEW

The last 10 minutes of this video are on identifying new plants.

∞ Video Link: Plant ID (38:50-46:03)

★ STUDENT TIP

Finding flowers this time of year may require purchasing or asking for old ones at local florists. Once you get used to the way keys work, you will be ready to start using keys to ID trees that are around in winter. If you have trouble identifying a tree, move on to another tree, get help from local experts, use your Tree in Winter book, search online for local winter tree ID help, or find a local naturalist to ask.

WEEK 1 45m Botany Lessons - Lesson 64

Flex Day

 Materials: Selected botany literature OR lab/field supplies

→ NOTE

Use this flex day to finish the week's reading from The Hidden Life of Trees or continue any lab/field activities. Suppose these activities are finished for the week. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as Growing a Revolution, or more active by pursuing a special study of interest.



Term 2

WEEK 1 ☐ 45m Botany Lessons - Lesson 65

Current Events

☐ Materials: Links

→ READ

Explore current events in Botany, keeping a weekly log or journal entry of your findings and thoughts in your Botany Notebook.

- ∞ Link: The Smithsonian
- ∞ Link: The Botanical Society of America
- ∞ Link: Botany at Science Daily
- ∞ Link: The Plant Press

OR

→ WRITE

Explore the intersection of science and faith, keeping a weekly log or journal entry of your findings and thoughts in your Botany Notebook.

- ∞ Link: Veritas Forum

WEEK 1 ☐ 60m Botany Labs - Lesson 13

☐ Materials: Elementary Studies in Plant Life

→ LAB DAY

Perform the practical work from Ch.9. Record your work in your notebook.

WEEK 2 ☐ 45m Botany Lessons - Lesson 66

Botany Literature

☐ Materials: The Hidden Life of Trees: The Illustrated Edition

→ NOTE

Take a few moments to look at the pictures on the pages you will read before beginning to read.

→ READ, NARRATE, & DISCUSS

Read about 14 pages per week; p.13-27.

→ WRITE

Write a narration in your Botany Notebook.

• PLAN WEEKLY

- ☐ Nature walk - look for the plants and/or features mentioned in your botany books, record observations; collect twigs and seeds
- ☐ science free read

WEEK 2 ☐ 45m Botany Lessons - Lesson 67

How Plants Work, Herbs in Winter

☐ Materials: Elementary Studies in Plant Life

→ READ, NARRATE, & DISCUSS

Ch.10

WEEK 2 ☐ 45m Botany Lessons - Lesson 68

Identification

☐ Materials: Botany in a Day

→ READ, NARRATE, & DISCUSS

p.22-25. Continue to familiarize yourself with the keys on p.26-35.



Term 2

WEEK 2 45m Botany Lessons - Lesson 69

Flex Day

Materials: Selected botany literature OR lab/field supplies

→ NOTE

Use this flex day to finish the week's reading from *The Hidden Life of Trees* or continue any lab/field activities. Suppose these activities are finished for the week. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as *Growing a Revolution*, or more active by pursuing a special study of interest.

WEEK 2 45m Botany Lessons - Lesson 70

Current Events

Materials: Links

→ READ/WRITE

Continue exploring the links from Week 1.

WEEK 2 60m Botany Labs - Lesson 14

Materials: Link

→ PLAN

Explore citizen science projects, such as Tree Snap, The Christmas Bird Count, BudBurst, and more. Choose one to do during the term that is related to trees or the creatures that depend on trees. Mark your calendar and obtain any supplies needed for the project. If you have another lab that week, you may shift your lab schedule to accommodate the project.

→ VIEW

∞ Link: SciStarter

WEEK 3 45m Botany Lessons - Lesson 71

Botany Literature

Materials: *The Hidden Life of Trees: The Illustrated Edition*

→ NOTE

Take a few moments to look at the pictures on the pages you will read before beginning to read.

→ READ, NARRATE, & DISCUSS

Read about 14 pages per week; p.27-40.

→ WRITE

Write a narration in your Botany Notebook.

• PLAN WEEKLY

- ☐ Nature walk - look for the plants and/or features mentioned in your botany books, record observations; collect twigs and seeds
- ☐ science free read

WEEK 3 45m Botany Lessons - Lesson 72

How Plants Work, Twigs

Materials: Elementary Studies in Plant Life, microscope, and prepared slides

→ READ, NARRATE, & DISCUSS

Ch.15

- Using a set of prepared slides and the linked procedure, look at some plant stems



Term 2

under the microscope. Record your work in your notebook.
∞ Link: Botanical Histology

→ VIEW

Watch this video on the internal anatomy of stems and roots.

∞ Video Link: Stem and Root Anatomy (11:53)

WEEK 3 45m Botany Lessons - Lesson 73

Identification

Materials: Link

→ VIEW

Watch this video on tree ID in winter for a few tips. Practice ID'ing trees.

∞ Video Link: Tree ID (15:33)

WEEK 3 45m Botany Lessons - Lesson 74

Flex Day

Materials: Selected botany literature OR lab/field supplies

→ NOTE

Use this flex day to finish the week's reading from *The Hidden Life of Trees* or continue any lab/field activities. Suppose these activities are finished for the week. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as *Growing a Revolution*, or more active by pursuing a special study of interest.

WEEK 3 45m Botany Lessons - Lesson 75

Current Events

Materials: Links

→ READ/WRITE

Continue exploring the links from Week 1.

WEEK 3 60m Botany Labs - Lesson 15

Materials: Elementary Studies in Plant Life

→ LAB DAY

Perform the practical work from Ch.15.

WEEK 4 45m Botany Lessons - Lesson 76

Botany Literature

Materials: *The Hidden Life of Trees: The Illustrated Edition*

→ NOTE

Take a few moments to look at the pictures on the pages you will read before beginning to read.

→ READ, NARRATE, & DISCUSS

Read about 14 pages per week; p.40-56.

→ WRITE

Write a narration in your Botany Notebook.

• PLAN WEEKLY

- ☐ Nature walk - look for the plants and/or features mentioned in your botany books, record observations; collect twigs and seeds
- ☐ science free read



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WEEK 4 45m Botany Lessons - Lesson 77

How Plants Work, Twigs

Materials: Links

→ PRACTICE

The two resources listed below are a very good introduction to how to use a key for identification. Study one or both to become comfortable with how they work. Then work to produce a twig key unique to your area. The twigs you have found and observed will be helpful.

→ VIEW

- ∞ Link: Winter Tree ID Resource (a printable key)
- ∞ Link: Winter Twig Key Example (a web-based key)

WEEK 4 45m Botany Lessons - Lesson 78

Identification

Materials: Botany in a Day, The Sibley Guide to Trees

→ READ

Read about the Fagaceae family on p.100 of Botany in a Day. Members of the 3 Beech family genera can be compared in The Sibley Guide on p.172-226. Which of these is most widespread in North America? What family characteristics do these different genera share? What distinguishes them? Find 2-3 species in the family found in your area to read more about.

WEEK 4 45m Botany Lessons - Lesson 79

Flex Day

Materials: Selected botany literature OR lab/field supplies

→ NOTE

Use this flex day to finish the week's reading from The Hidden Life of Trees or continue any lab/field activities. Suppose these activities are finished for the week. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as Growing a Revolution, or more active by pursuing a special study of interest.

WEEK 4 45m Botany Lessons - Lesson 80

Current Events

Materials: Links

→ READ/WRITE

Continue exploring the links from Week 1.

WEEK 4 60m Botany Labs - Lesson 16

Materials: Sibley Guide to Trees

→ READ, NARRATE, & DISCUSS

"Twigs" and "Buds" p.xxxv-xxxvii

→ PRACTICE

Identify the features described among the twigs you have collected.



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WEEK 5 45m Botany Lessons - Lesson 81

Botany Literature

Materials: The Hidden Life of Trees: The Illustrated Edition

→ NOTE

Take a few moments to look at the pictures on the pages you will read before beginning to read.

→ READ, NARRATE, & DISCUSS

Read about 14 pages per week; p.56-68.

→ WRITE

Write a narration in your Botany Notebook.

• PLAN WEEKLY

- ☐ Nature walk - look for the plants and/or features mentioned in your botany books, record observations; collect twigs and seeds
- ☐ science free read

WEEK 5 45m Botany Lessons - Lesson 82

How Plants Work, Twigs

→ PRACTICE

Work on your twig key.

WEEK 5 45m Botany Lessons - Lesson 83

Identification

Materials: Botany in a Day, The Sibley Guide to Trees

→ READ

Read about the Betulaceae family on p.101 of Botany in a Day. Members of the 5 Birch family genera can be compared in The Sibley Guide on p.150-171. What family characteristics do these different genera share? What distinguishes them? Find 2-3 species in the family found in your area to read more about. Are there even more species in your local tree guide or on your local cooperative extension website?

WEEK 5 45m Botany Lessons - Lesson 84

Flex Day

Materials: Selected botany literature OR lab/field supplies

→ NOTE

Use this flex day to finish the week's reading from The Hidden Life of Trees or continue any lab/field activities. Suppose these activities are finished for the week. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as Growing a Revolution, or more active by pursuing a special study of interest.

WEEK 5 45m Botany Lessons - Lesson 85

Current Events

Materials: Links

→ READ/WRITE

Continue exploring the links from Week 1.



Term 2

WEEK 5 ☐ 60m Botany Labs - Lesson 17

☐ Materials: Sibley Guide to Trees

→ READ, NARRATE, & DISCUSS

"Bark" p.xxxvii-xxxviii

→ PRACTICE

Identify the features described among various trees.

WEEK 6 ☐ 45m Botany Lessons - Lesson 86

Botany Literature

☐ Materials: The Hidden Life of Trees: The Illustrated Edition

→ NOTE

Take a few moments to look at the pictures on the pages you will read before beginning to read.

→ READ, NARRATE, & DISCUSS

Read about 14 pages per week; p.68-81.

→ WRITE

Write a narration in your Botany Notebook.

• PLAN WEEKLY

☐ Nature walk - look for the plants and/or features mentioned in your botany books, record observations; collect twigs and seeds

☐ science free read

WEEK 6 ☐ 45m Botany Lessons - Lesson 87

How Plants Work, Twigs

→ PRACTICE

Work on your twig key.

WEEK 6 ☐ 45m Botany Lessons - Lesson 88

Identification

☐ Materials: Botany in a Day, The Sibley Guide to Trees

→ READ

Read about the Sapindaceae family on p.102-103 of Botany in a Day. Members of 2 Soapberry family genera can be compared in The Sibley Guide p.324-351. What family characteristics do these genera share? What distinguishes them? Find 2-3 species in the family located in your area to read more about. Are there even more species in your local tree guide or on your local cooperative extension website?

WEEK 6 ☐ 45m Botany Lessons - Lesson 89

Flex Day

☐ Materials: Selected botany literature OR lab/field supplies

→ NOTE

Use this flex day to finish the week's reading from The Hidden Life of Trees or continue any lab/field activities. Suppose these activities are finished for the week. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as Growing a Revolution, or more active by pursuing a special study of interest.



Term 2

WEEK 6 ☐ 45m Botany Lessons - Lesson 90

Current Events

☐ Materials: Links

→ READ/WRITE

Continue exploring the links from Week 1.

WEEK 6 ☐ 60m Botany Labs - Lesson 18

→ PRACTICE

Catch-up day to accommodate citizen science.

WEEK 7 ☐ 45m Botany Lessons - Lesson 91

Botany Literature

☐ Materials: The Hidden Life of Trees: The Illustrated Edition

→ NOTE

Take a few moments to look at the pictures on the pages you will read before beginning to read.

→ READ, NARRATE, & DISCUSS

Read about 14 pages per week; p.83-98.

→ WRITE

Write a narration in your Botany Notebook.

• PLAN WEEKLY

- ☐ Nature walk -look for the plants and/or features mentioned in your botany books, record observations; collect twigs and seeds
- ☐ science free read

WEEK 7 ☐ 45m Botany Lessons - Lesson 92

How Plants Work, Fruits and Seeds

☐ Materials: Elementary Studies in Plant Life

→ READ, NARRATE, & DISCUSS

Ch.24

WEEK 7 ☐ 45m Botany Lessons - Lesson 93

Identification

☐ Materials: Botany in a Day, The Sibley Guide to Trees

→ READ

Read about the Pinaceae family on p.46 of Botany in a Day. Compare members of the Pine family in The Sibley Guide p.4-59. What family characteristics do these different genera share? What distinguishes them? Find 2-3 species in the family found in your area to read more about. Are there even more species in your local tree guide or on your local cooperative extension website?

WEEK 7 ☐ 45m Botany Lessons - Lesson 94

Flex Day

☐ Materials: Selected botany literature OR lab/field supplies

→ NOTE

Use this flex day to finish the week's reading from The Hidden Life of Trees or continue any lab/field activities. Suppose these activities are finished for the week. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as Growing a Revolution, or more active by pursuing a special study of interest.



Term 2

WEEK 7 45m Botany Lessons - Lesson 95

Current Events

Materials: Links

→ READ/WRITE

Continue exploring the links from Week 1.

WEEK 7 60m Botany Labs - Lesson 19

→ PRACTICE

Catch-up day to accommodate citizen science. Or, if you are ready, move on to next week's project!

WEEK 8 45m Botany Lessons - Lesson 96

Botany Literature

Materials: The Hidden Life of Trees: The Illustrated Edition

→ NOTE

Take a few moments to look at the pictures on the pages you will read before beginning to read.

→ READ, NARRATE, & DISCUSS

Read about 14 pages per week; p.98-108.

→ WRITE

Write a narration in your Botany Notebook.

• PLAN WEEKLY

- ☐ Nature walk -record observations
- ☐ science free read

WEEK 8 45m Botany Lessons - Lesson 97

How Plants Work, How Fruits and Seeds Spread

Materials: Elementary Studies in Plant Life

→ READ, NARRATE, & DISCUSS

Ch.25

WEEK 8 45m Botany Lessons - Lesson 98

Identification

Materials: Botany in a Day, The Sibley Guide to Trees

→ READ

Read about the Magnolia (p.52) and Juglandaceae (p.99) families in Botany in a Day. Members of the Magnolia family can be compared in The Sibley Guide p.92-100. Members of the Walnut family can be compared on p.136-149.

OR

Read about the Cupressaceae family on p.48 of Botany in a Day. Members of the Cypress family can be compared in The Sibley Guide p.60-77.

- Find 2-3 species in the family located in your area to read more about. Are there even more species in your local tree guide or on your local cooperative extension website?



Term 2

WEEK 8 45m Botany Lessons - Lesson 99

Flex Day

Materials: Selected botany literature OR lab/field supplies

→ NOTE

Use this flex day to finish the week's reading from *The Hidden Life of Trees* or continue any lab/field activities. Suppose these activities are finished for the week. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as *Growing a Revolution*, or more active by pursuing a special study of interest.

WEEK 8 45m Botany Lessons - Lesson 100

Current Events

Materials: Links

→ READ/WRITE

Continue exploring the links from Week 1.

WEEK 8 60m Botany Labs - Lesson 20

Materials: Seed collection or ethnobotany project, student choice

→ PRACTICE

For the last several weeks of the term, you will work on 1 of 2 projects:

1. Create a "seed bank" to preserve and present your collection to your teacher; or
2. Choose one of your local plants, and research how the plant was used culturally by a group of people. Present your findings to your teacher. The video link below contains a great example of ethnobotany.

→ VIEW

∞ Video Link: Ethnobotany (6:09)

Decide today which project you will do and what form it will take. A seed bank could be a collection of paintings, models/sculptures, poems, musical motifs, etc.

Ethnobotany research could be presented as a brochure designed on Canva, a slide/poster presentation, a fireside tale, etc. Once you are ready, begin working!

WEEK 9 45m Botany Lessons - Lesson 101

Botany Literature

Materials: *The Hidden Life of Trees: The Illustrated Edition*

→ NOTE

Take a few moments to look at the pictures on the pages you will read before beginning to read.

→ READ, NARRATE, & DISCUSS

Read about 14 pages per week; p.108-123.

→ WRITE

Write a narration in your Botany Notebook.

• PLAN WEEKLY

- ☐ Nature walk - record observations
- ☐ science free read



Term 2

WEEK 9 45m Botany Lessons - Lesson 102

How Plants Work, Fruits and Seeds

Materials: The Sibley Guide to Trees

→ READ, NARRATE, & DISCUSS

"Fruit" p.xxxiii

→ PRACTICE

Begin next week's practice, if time remains.

WEEK 9 45m Botany Lessons - Lesson 103

Identification

Materials: The Sibley Guide to Trees

→ READ

Consider a tree or shrub in your area that has not been studied yet this term. Use the Table of Contents in the front or the Index in the back of the Sibley Guide to find and read about its family. What characteristics do the different genera share? What distinguishes them? Read about at least 2 other family members. Are there even more species in your local tree guide or on your local cooperative extension website?

WEEK 9 45m Botany Lessons - Lesson 104

Flex Day

Materials: Selected botany literature OR lab/field supplies

→ NOTE

Use this flex day to finish the week's reading from *The Hidden Life of Trees* or continue any lab/field activities. Suppose these activities are finished for the week. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as *Growing a Revolution*, or more active by pursuing a special study of interest.

WEEK 9 45m Botany Lessons - Lesson 105

Current Events

Materials: Links

→ READ/WRITE

Continue exploring the links from Week 1.

WEEK 9 60m Botany Labs - Lesson 21

Materials: Seed collection or ethnobotany project, student choice

→ PRACTICE

Work on your project.

WEEK 10 45m Botany Lessons - Lesson 106

Botany Literature

Materials: *The Hidden Life of Trees: The Illustrated Edition*

→ NOTE

Take a few moments to look at the pictures on the pages you will read before beginning to read.

• PLAN WEEKLY

- ☐ Nature walk - record observations
- ☐ science free read



Term 2

→ READ, NARRATE, & DISCUSS

Read about 14 pages per week; p.125-140.

→ WRITE

Write a narration in your Botany Notebook.

WEEK 10 45m Botany Lessons - Lesson 107

How Plants Work, Fruits and Seeds

Materials: The Sibley Guide to Trees

→ PRACTICE

Observe your collection of seeds. Can you categorize them according to the different types described in your guide? Can you identify the plant from which each one came? What type of habitat does the parent plant live in? Or in what habitat did you find the seed? Based on the features of the seed, what is its mechanism of spreading? How do you suppose this benefits the plant? You will have more time for this observation next week.

WEEK 10 45m Botany Lessons - Lesson 108

Identification

Materials: The Sibley Guide to Trees

→ READ

Look through the Table of Contents in the Sibley Guide and find a tree family that sounds interesting. Read about that family. What characteristics are unique to this family? Are there any members of this family near you? You can search your local cooperative extension if you don't see any in the Guide.

WEEK 10 45m Botany Lessons - Lesson 109

Flex Day

Materials: Selected botany literature OR lab/field supplies

→ NOTE

Use this flex day to finish the week's reading from *The Hidden Life of Trees* or continue any lab/field activities. Suppose these activities are finished for the week. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as *Growing a Revolution*, or more active by pursuing a special study of interest.

WEEK 10 45m Botany Lessons - Lesson 110

Current Events

Materials: Links

→ READ/WRITE

Continue exploring the links from Week 1.

WEEK 10 60m Botany Labs - Lesson 22

Materials: Seed collection or ethnobotany project, student choice

→ PRACTICE

Work on your project.



Term 2

WEEK 11 45m Botany Lessons - Lesson 111

Botany Literature

Materials: The Hidden Life of Trees: The Illustrated Edition

→ NOTE

Take a few moments to look at the pictures on the pages you will read before beginning to read.

→ READ, NARRATE, & DISCUSS

Read about 14 pages per week; p.140-158.

→ WRITE

Write a narration in your Botany Notebook.

• PLAN WEEKLY

- ☐ Nature walk - record observations
- ☐ science free read

WEEK 11 45m Botany Lessons - Lesson 112

How Plants Work, Fruits and Seeds

Materials: The Sibley Guide to Trees

→ PRACTICE

Observe your collection of seeds. Can you categorize them according to the different types described in your guide? Can you identify the plant from which each one came? What type of habitat does the parent plant live in? Or in what habitat did you find the seed? Based on the features of the seed, what is its mechanism of spreading? How do you suppose this benefits the plant?

WEEK 11 45m Botany Lessons - Lesson 113

Identification

→ OBSERVE

Practice ID'ing trees. Record your observations.

WEEK 11 45m Botany Lessons - Lesson 114

Flex Day

Materials: Selected botany literature OR lab/field supplies

→ NOTE

Use this flex day to finish the week's reading from The Hidden Life of Trees or continue any lab/field activities. Suppose these activities are finished for the week. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as Growing a Revolution, or more active by pursuing a special study of interest.

WEEK 11 45m Botany Lessons - Lesson 115

Current Events

Materials: Links

→ READ/WRITE

Continue exploring the links from Week 1.



Term 2

WEEK 11 60m Botany Labs - Lesson 23

Materials: Seed collection or ethnobotany project, student choice

→ PRACTICE

Work on your project.

WEEK 12 45m Botany Lessons - Lesson 116

Exam Week

→ EXAMS

Answer question(s) related to the course.

WEEK 12 45m Botany Lessons - Lesson 117

Exam Week

→ EXAMS

Answer question(s) related to the course.

WEEK 12 45m Botany Lessons - Lesson 118

Exam Week

→ EXAMS

Answer question(s) related to the course.

WEEK 12 45m Botany Lessons - Lesson 119

Exam Week

→ EXAMS

Answer question(s) related to the course.

WEEK 12 45m Botany Lessons - Lesson 120

Exam Week

→ EXAMS

Answer question(s) related to the course.

WEEK 12 60m Botany Labs - Lesson 24

Exam Week

→ EXAMS

Answer question(s) related to the course.



Term 3

WEEK 1 45m Botany Lessons - Lesson 121

Botany Literature

Materials: Oaxaca Journal

→ INTRO

The lesson plans give you an idea of how many pages to read each week to finish the book by the end of the term. If one day is not enough, then you might read about half on one day and half on the flex day. If one day is enough, then decide how you would like to use your flex day. Consider the ideas provided on Day 4 of the week.

→ READ, NARRATE, & DISCUSS

Read about 14.5 pages per week, to p.15.

→ DRAW

Make a large sketch of the region where the book takes place. A nice map is on the first page. Then, as you read, fill in interesting things that he describes. At the end, you will have a map that represents some of the mental images that you have of the area after reading his descriptive writing.

• PLAN WEEKLY

- ☐ Nature walk -record observations
- ☐ science free read

WEEK 1 45m Botany Lessons - Lesson 122

How Plants Work, Germination Part 1

Materials: Elementary Studies in Plant Life

→ READ, NARRATE, & DISCUSS

Ch.11

→ REFLECT

If you have already taken Biology, recall what you learned about why oxygen is necessary for cellular respiration. Reflect on why you think the author did not explain all the details in the process.

WEEK 1 45m Botany Lessons - Lesson 123

Identification

Materials: Botany in a Day

→ INTRO

For a study of Botany, springtime is the time of year we have been waiting for! There are a lot more flower families to look for outside!

→ READ

Read about and look for Ranunculaceae (p.55) and Salicaceae (p.76) families.

WEEK 1 45m Botany Lessons - Lesson 124

Flex Day

Materials: Selected botany literature OR lab/field supplies

→ NOTE

Use this flex day to finish the week's reading from Oaxaca Journal or continue any lab/field activities. Suppose these activities are finished for the week. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as *Growing a Revolution*, or more active by pursuing a special study of interest.



Term 3

WEEK 1 45m Botany Lessons - Lesson 125

Current Events

Materials: Links

→ READ

Explore current events in botany, keeping a weekly log or journal entry of your findings and thoughts in your Botany Notebook.

- ∞ Link: The Smithsonian
- ∞ Link: The Botanical Society of America
- ∞ Link: Botany at Science Daily
- ∞ Link: The Plant Press

OR

→ WRITE

Explore the intersection of science and faith, keeping a weekly log or journal entry of your findings and thoughts in your Botany Notebook.

- ∞ Link: Veritas Forum

WEEK 1 60m Botany Labs - Lesson 25

Materials: Elementary Studies in Plant Life or link

→ LAB DAY

Perform selected practical work from Ch.11-12. Record your work in your notebook.

OR

→ PLAN

Choose another botany-related citizen science project for spring.

→ VIEW

- ∞ Link: SciStarter

WEEK 2 45m Botany Lessons - Lesson 126

Botany Literature

Materials: Oaxaca Journal

→ READ, NARRATE, & DISCUSS

Read about 14.5 pages per week; p.15-30.

→ MAP WORK

Add to your sketch map.

• PLAN WEEKLY

- ☐ Nature walk - record observations
- ☐ science free read

WEEK 2 45m Botany Lessons - Lesson 127

How Plants Work, Germination Part 2

Materials: Elementary Studies in Plant Life

→ READ, NARRATE, & DISCUSS

Ch.12



Term 3

WEEK 2 45m Botany Lessons - Lesson 128

Identification

Materials: Botany in a Day

→ READ

Read about and look for Sapindaceae (p.102) and Cornaceae (p.124) families.

WEEK 2 45m Botany Lessons - Lesson 129

Flex Day

Materials: Selected botany literature OR lab/field supplies

→ NOTE

Use this flex day to finish the week's reading from Oaxaca Journal or continue any lab/field activities. Suppose these activities are finished for the week. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as Growing a Revolution, or more active by pursuing a special study of interest.

WEEK 2 45m Botany Lessons - Lesson 130

Current Events

Materials: Links

→ READ/WRITE

Continue exploring the links from Week 1.

WEEK 2 60m Botany Labs - Lesson 26

Materials: Elementary Studies in Plant Life

→ LAB DAY

Perform selected practical work from Ch.11-12. Record your work in your notebook.

WEEK 3 45m Botany Lessons - Lesson 131

Botany Literature

Materials: Oaxaca Journal

→ READ, NARRATE, & DISCUSS

Read about 14.5 pages per week; p.30-44.

→ MAP WORK

Add to your sketch map.

• PLAN WEEKLY

- ☐ Nature walk -record observations
- ☐ science free read

WEEK 3 45m Botany Lessons - Lesson 132

How Plants Work, Herbs in Spring

Materials: Elementary Studies in Plant Life

→ READ, NARRATE, & DISCUSS

Ch.13

→ PRACTICE

Pick a question at the end of the chapter to answer fully in your notebook to serve as your narration.



Term 3

WEEK 3 45m Botany Lessons - Lesson 133

Identification

Materials: Botany in a Day

→ READ

Read about and look for the Iridaceae (p.201) family. Compare it to the Liliaceae family that you already know about and figure out the difference.

WEEK 3 45m Botany Lessons - Lesson 134

Flex Day

Materials: Selected botany literature OR lab/field supplies

→ NOTE

Use this flex day to finish the week's reading from Oaxaca Journal or continue any lab/field activities. Suppose these activities are finished for the week. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as Growing a Revolution, or more active by pursuing a special study of interest.

WEEK 3 45m Botany Lessons - Lesson 135

Current Events

Materials: Links

→ READ/WRITE

Continue exploring the links from Week 1.

WEEK 3 60m Botany Labs - Lesson 27

Materials: Elementary Studies in Plant Life

→ LAB DAY

Perform the practical work from Ch.13. Record your work in your notebook.

OR

→ PRACTICE

Catch-up day to accommodate citizen science.

WEEK 4 45m Botany Lessons - Lesson 136

Botany Literature

Materials: Oaxaca Journal

→ READ, NARRATE, & DISCUSS

Read about 14.5 pages per week; p.44-56.

→ MAP WORK

Add to your sketch map.

• PLAN WEEKLY

- ☐ Nature walk - record observations
- ☐ science free read



Term 3

WEEK 4 45m Botany Lessons - Lesson 137
How Plants Work, Trees and Shrubs in Spring

Materials: Elementary Studies in Plant Life

→ READ, NARRATE, & DISCUSS
Ch.14

WEEK 4 45m Botany Lessons - Lesson 138
Identification

Materials: Botany in a Day

→ READ
Read about and look for Violaceae (p.75) and Caryophyllaceae (p.114) families.

WEEK 4 45m Botany Lessons - Lesson 139
Flex Day

Materials: Selected botany literature OR lab/field supplies

→ NOTE
Use this flex day to finish the week's reading from Oaxaca Journal or continue any lab/field activities. Suppose these activities are finished for the week. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as *Growing a Revolution*, or more active by pursuing a special study of interest.

WEEK 4 45m Botany Lessons - Lesson 140
Current Events

Materials: Links

→ READ/WRITE
Continue exploring the links from Week 1.

WEEK 4 60m Botany Labs - Lesson 28

Materials: Elementary Studies in Plant Life

→ LAB DAY
Perform the practical work from Ch.14. If you would like to try Question 1, you may look for green ash, white ash, or walnut instead of horse chestnut. Record your work in your notebook.

WEEK 5 45m Botany Lessons - Lesson 141
Botany Literature

Materials: Oaxaca Journal

→ READ, NARRATE, & DISCUSS
Read about 14.5 pages per week; p.56-70.

→ MAP WORK
Add to your sketch map.

- PLAN WEEKLY
- ☐ Nature walk - record observations
- ☐ science free read



Term 3

WEEK 5

45m Botany Lessons - Lesson 142

How Plants Work, Alternative Ways of Obtaining Food

Materials: Elementary Studies in Plant Life

→ READ, NARRATE, & DISCUSS
Ch.18

→ PRACTICE

Pick a question at the end of the chapter to answer fully in your notebook to serve as your narration.

WEEK 5

45m Botany Lessons - Lesson 143

Identification

Materials: Botany in a Day

→ READ

Read about and look for the Arecaceae (p.184) family.

WEEK 5

45m Botany Lessons - Lesson 144

Flex Day

Materials: Selected botany literature OR lab/field supplies

→ NOTE

Use this flex day to finish the week's reading from Oaxaca Journal or continue any lab/field activities. Suppose these activities are finished for the week. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as Growing a Revolution, or more active by pursuing a special study of interest.

WEEK 5

45m Botany Lessons - Lesson 145

Current Events

Materials: Links

→ READ/WRITE

Continue exploring the links from Week 1.

WEEK 5

60m Botany Labs - Lesson 29

Materials: Botany in a Day

→ OBSERVE

As spring arrives, try to find an example of each family described in Botany in a Day. Record your observations.

WEEK 6

45m Botany Lessons - Lesson 146

Botany Literature

Materials: Oaxaca Journal

→ READ, NARRATE, & DISCUSS

Read about 14.5 pages per week; p.70-85.

→ MAP WORK

Add to your sketch map.

• PLAN WEEKLY

- ☐ Nature walk -record observations
- ☐ science free read



Term 3

WEEK 6 45m Botany Lessons - Lesson 147

How Plants Work, Growth

Materials: Elementary Studies in Plant Life

→ READ, NARRATE, & DISCUSS
Ch.19

→ PRACTICE
Pick a question at the end of the chapter to answer fully in your notebook to serve as your narration.

WEEK 6 45m Botany Lessons - Lesson 148

Identification

→ OBSERVE
Spend the time outside looking at seeds, trees, and plants, recording your observations. You should be able to see so much more now than when you started in the fall!

WEEK 6 45m Botany Lessons - Lesson 149

Flex Day

Materials: Selected botany literature OR lab/field supplies

→ NOTE
Use this flex day to finish the week's reading from Oaxaca Journal or continue any lab/field activities. Suppose these activities are finished for the week. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as *Growing a Revolution*, or more active by pursuing a special study of interest.

WEEK 6 45m Botany Lessons - Lesson 150

Current Events

Materials: Links

→ READ/WRITE
Continue exploring the links from Week 1.

WEEK 6 60m Botany Labs - Lesson 30

Materials: Elementary Studies in Plant Life

→ LAB DAY
Perform the practical work from Ch.19. Record your work in your notebook.

WEEK 7 45m Botany Lessons - Lesson 151

Botany Literature

Materials: Oaxaca Journal

→ READ, NARRATE, & DISCUSS
Read about 14.5 pages per week; p.89-100.

→ MAP WORK
Add to your sketch map.

- PLAN WEEKLY
- ☐ Nature walk -record observations
- ☐ science free read



Term 3

WEEK 7 45m Botany Lessons - Lesson 152

How Plants Work, Plants in Dry Places

Materials: Elementary Studies in Plant Life

→ READ, NARRATE, & DISCUSS
Ch.20

→ PRACTICE
Pick a question at the end of the chapter to answer fully in your notebook to serve as your narration.

WEEK 7 45m Botany Lessons - Lesson 153

Identification

→ OBSERVE
Spend the time outside looking at seeds, trees, and plants, recording your observations. You should be able to see so much more now than when you started in the fall!

WEEK 7 45m Botany Lessons - Lesson 154

Flex Day

Materials: Selected botany literature OR lab/field supplies

→ NOTE
Use this flex day to finish the week's reading from Oaxaca Journal or continue any lab/field activities. Suppose these activities are finished for the week. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as *Growing a Revolution*, or more active by pursuing a special study of interest.

WEEK 7 45m Botany Lessons - Lesson 155

Current Events

Materials: Links

→ READ/WRITE
Continue exploring the links from Week 1.

WEEK 7 60m Botany Labs - Lesson 31

Materials: Elementary Studies in Plant Life

→ LAB DAY
Perform selected practical work from Ch.20. Record your work in your notebook.

WEEK 8 45m Botany Lessons - Lesson 156

Botany Literature

Materials: Oaxaca Journal

→ READ, NARRATE, & DISCUSS
Read about 14.5 pages per week; p.103-114.

→ MAP WORK
Add to your sketch map.

- PLAN WEEKLY
 - ☐ Nature walk -record observations
 - ☐ science free read



Term 3

WEEK 8 45m Botany Lessons - Lesson 157

How Plants Work, Plants in Wet Places

Materials: Elementary Studies in Plant Life

→ READ, NARRATE, & DISCUSS

Read the first half of Ch.21.

→ PRACTICE

Pick a question at the end of the chapter to answer fully in your notebook to serve as your narration.

WEEK 8 45m Botany Lessons - Lesson 158

Identification

→ OBSERVE

Spend the time outside looking at seeds, trees, and plants, recording your observations. You should be able to see so much more now than when you started in the fall!

WEEK 8 45m Botany Lessons - Lesson 159

Flex Day

Materials: Selected botany literature OR lab/field supplies

→ NOTE

Use this flex day to finish the week's reading from Oaxaca Journal or continue any lab/field activities. Suppose these activities are finished for the week. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as *Growing a Revolution*, or more active by pursuing a special study of interest.

WEEK 8 45m Botany Lessons - Lesson 160

Current Events

Materials: Links

→ READ/WRITE

Continue exploring the links from Week 1.

WEEK 8 60m Botany Labs - Lesson 32

Materials: Link, household items indicated, microscope, and glass slides

→ LAB DAY

Return to one of your soil sampling locations from Term 1. Use the procedure at the link to evaluate microbial life in soil from at least 3 locations and compare to your observations from Term 1.

→ VIEW

∞ Link: Soil Sampling Procedure



Term 3

WEEK 9 45m Botany Lessons - Lesson 161

Botany Literature

Materials: Oaxaca Journal

→ READ, NARRATE, & DISCUSS

Read about 14.5 pages per week; p.114-127.

→ MAP WORK

Add to your sketch map.

• PLAN WEEKLY

Nature walk - record observations

science free read

WEEK 9 45m Botany Lessons - Lesson 162

How Plants Work, Plants in Wet Places

Materials: Elementary Studies in Plant Life

→ READ, NARRATE, & DISCUSS

Read the second half of Ch.21.

→ PRACTICE

Pick a question at the end of the chapter to answer fully in your notebook to serve as your narration.

WEEK 9 45m Botany Lessons - Lesson 163

Identification

→ OBSERVE

Spend the time outside looking at seeds, trees, and plants, recording your observations. You should be able to see so much more now than when you started in the fall!

WEEK 9 45m Botany Lessons - Lesson 164

Flex Day

Materials: Selected botany literature OR lab/field supplies

→ NOTE

Use this flex day to finish the week's reading from Oaxaca Journal or continue any lab/field activities. Suppose these activities are finished for the week. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as *Growing a Revolution*, or more active by pursuing a special study of interest.

WEEK 9 45m Botany Lessons - Lesson 165

Current Events

Materials: Links

→ READ/WRITE

Continue exploring the links from Week 1.

WEEK 9 60m Botany Labs - Lesson 33

Materials: Elementary Studies in Plant Life

→ LAB DAY

Perform selected practical work from Ch.21. Record your work in your notebook.



Term 3

WEEK 10 45m Botany Lessons - Lesson 166

Botany Literature

Materials: Oaxaca Journal

→ READ, NARRATE, & DISCUSS

Read about 14.5 pages per week; p.127-140.

→ MAP WORK

Add to your sketch map.

• PLAN WEEKLY

Nature walk - record observations

science free read

WEEK 10 45m Botany Lessons - Lesson 167

How Plants Work, Flowering Parts

Materials: Elementary Studies in Plant Life

→ READ, NARRATE, & DISCUSS

Ch.22

→ PRACTICE

Pick a question at the end of the chapter to answer fully in your notebook to serve as your narration.

WEEK 10 45m Botany Lessons - Lesson 168

Identification

→ OBSERVE

Spend the time outside looking at seeds, trees, and plants, recording your observations. You should be able to see so much more now than when you started in the fall!

WEEK 10 45m Botany Lessons - Lesson 169

Flex Day

Materials: Selected botany literature OR lab/field supplies

→ NOTE

Use this flex day to finish the week's reading from Oaxaca Journal or continue any lab/field activities. Suppose these activities are finished for the week. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as *Growing a Revolution*, or more active by pursuing a special study of interest.

WEEK 10 45m Botany Lessons - Lesson 170

Current Events

Materials: Links

→ READ/WRITE

Continue exploring the links from Week 1.

WEEK 10 60m Botany Labs - Lesson 34

Materials: Botany in a Day

→ OBSERVE

As spring arrives, try to find an example of each family described in *Botany in a Day*. Record your observations.



Term 3

WEEK 11 45m Botany Lessons - Lesson 171

Botany Literature

Materials: Oaxaca Journal

→ READ, NARRATE, & DISCUSS

Read about 14.5 pages per week; p.141-159.

→ MAP WORK

Add to your sketch map.

• PLAN WEEKLY

Nature walk - record observations

science free read

WEEK 11 45m Botany Lessons - Lesson 172

How Plants Work

Materials: Links

→ VIEW

Watch these videos on stomata. Then take a few moments to observe the new leaves on the trees. How do they compare to those observed in Term 1?

∞ Video Link: Inside a Leaf (2:52)

∞ Video Link: Stomata #2 (3:50)

WEEK 11 45m Botany Lessons - Lesson 173

Identification

→ OBSERVE

Spend the time outside looking at seeds, trees, and plants, recording your observations. You should be able to see so much more now than when you started in the fall!

WEEK 11 45m Botany Lessons - Lesson 174

Flex Day

Materials: Selected botany literature OR lab/field supplies

→ NOTE

Use this flex day to finish the week's reading from Oaxaca Journal or continue any lab/field activities. Suppose these activities are finished for the week. In that case, you may use this day to tailor the course to be more literary by reading an additional text, such as *Growing a Revolution*, or more active by pursuing a special study of interest.

WEEK 11 45m Botany Lessons - Lesson 175

Current Events

Materials: Links

→ READ/WRITE

Continue exploring the links from Week 1.

WEEK 11 60m Botany Labs - Lesson 35

→ CATCH UP DAY

Use this time to catch up in this or any other course.



Term 3

WEEK 12 45m Botany Lessons - Lesson 176

Exam Week

→ EXAMS

Answer question(s) related to the course.

• PLAN WEEKLY

☐ Nature walk - record observations

☐ science free read

WEEK 12 45m Botany Lessons - Lesson 177

Exam Week

→ EXAMS

Answer question(s) related to the course.

WEEK 12 45m Botany Lessons - Lesson 178

Exam Week

→ EXAMS

Answer question(s) related to the course.

WEEK 12 45m Botany Lessons - Lesson 179

Exam Week

→ EXAMS

Answer question(s) related to the course.

WEEK 12 45m Botany Lessons - Lesson 180

Exam Week

→ EXAMS

Answer question(s) related to the course.

WEEK 12 60m Botany Labs - Lesson 36

Exam Week

→ EXAMS

Answer question(s) related to the course.

Science: Botany

Examination

Term 1

Botany Lessons

• Day 1:

- What a Plant Knows: Pick one aspect of things that a "plant knows" and write about it.

• Day 2:

- Elementary Studies: Explain what you know about photosynthesis. Use sketches or diagrams.
- Elementary Studies: What are the different parts of a flower, and what are they used for? Use sketches or diagrams.

• Day 3:

- Botany in a Day: Outline with words and sketches the 8 common plant families we studied and how to recognize them.

• Day 4:

- The Milkweed Lands: Write a summary OR draw/diagram a 4-panel illustration of the ecological role that milkweed plays over the seasons.

• Day 5:

- Practical work or links: Describe with words and sketches your observations of 2 local plants OR discuss the importance of soil quality.

Term 2

Botany Lessons

• Day 1:

- The Hidden Life of Trees: Pick one aspect of The Hidden Life of Trees and write about it.

• Day 2:

- Elementary Studies and/or practical work: Use words or pictures to contrast the form and function of at least two types of seeds OR describe the process of keying out a flower, twig, or tree.

• Day 3:

- Practical work: Use words and sketches to describe your observations of 2 local trees.

• Day 4:

- Current events: Discuss one current event in botany that you read about this term.

• Day 5:

- Practical work: Present your citizen science project to your teacher.

Term 3

Botany Lessons

· Day 1:

- Oaxaca Journal: Describe some of the ferns the author saw in Oaxaca and some of his observations.

· Day 2:

- Elementary Studies: Describe the process of germination. How are a corn seed and a bean seed different?

· Day 3:

- Botany in a Day and practical work: What new plants have you seen locally since spring has arrived? How many of the 8 first families that we learned have you found?

· Day 4:

- Botany in a Day and practical work: Describe how you prepared soil samples for microscopic examination and explain any observations you made between different locations or times of year.

· Day 5:

- Practical work: Show your teacher your Botany Notebook. Select a favorite page to tell about and scan.